MIL-C-16671B
15 September 1976
SUPERSEDING
MIL-C-16671A
25 October 1967

MILITARY SPECIFICATION

CRUTCH, ALUMINUM, CANE TYPE, ADJUSTABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of adjustable cane type aluminum crutch.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-A-225/6	- Aluminum Alloy Bar, Rod, and Wire; Rolled, Drawn, or Cold Finished, 2024.
QQ-A-225/8	- Aluminum Alloy Bar, Rod, Wire and Special Shapes, Rolled, Drawn, or Cold Finished, 6061.
QQ-A-250/8	- Aluminum Alloy, 5052 Plate and Sheet.
QQ-A-367	- Aluminum Alloy Forgings.
QQ-A-591	- Aluminum Alloy Die Castings.
QQ-A-601	- Aluminum Alloy Sand Castings.
QQ-S-766	- Steel Plate, Sheet, and Strip Corrosion-Resisting.
WW-T-700/3	- Tube, Aluminum Alloy, Drawn, Seamless 2024.
PPP-B-585	- Box, Wood, Wirebound.
PPP-B-601	- Box, Wood, Cleated-Plywood.
PPP-B-621	- Box, Wood, Nailed, and Lock-Corner.
PPP-B-636	- Box, Shipping, Fiberboard.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq., Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, Code ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6515

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MIL-A-8625	- Anodic Coatings, for Aluminum and Aluminum Alloys.
MIL-C-5541	- Chemical Conversion Coating on Aluminum and Aluminum Alloys.
MIL-L-10547	- Liners, Case and Sheet, Overwrap, Water-Vaporproof or Waterproof, Flexible.

STANDARDS

FEDERAL

FED-STD-151	- Metals; Test Methods.
FED-STD-601	- Rubber: Sampling and Testing.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

DRAWING

Defense Personnel Support Center

1367 CRUTCH, CANE TYPE, ADJUSTABLE.

(Copies of specifications, standards, drawing, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 412	Tension Testing of Vulcanized Rubber.
D 570	Test for Water Absorption of Plastics.
D 635	Test for Rate of Burning or Extent and Time of Burning, or Both, of Self-Supporting Plastics.
D 638	Test for Tensile Properties of Plastics.

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D 792 Tests for Specific Gravity and Density of Plastics by Displacement.
D 2240 Test for Indentation Hardness of Rubber and Plastics by Means of a Durometer.

(Application for copies should be addressed to American Society for Testing and Materials, 1903 Race Street, Philadelphia, PA 19103).

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Materials. Materials shall conform to the applicable portions of the specifications, standards, and drawing specified herein. When materials to be used are not specifically designated, those used shall be of the highest commercial quality and entirely suitable for the purpose intended.

3.1.1 Tubing. The tubing used for the crutch shall be drawn seamless aluminum conforming to 2024-T3 of WW-T-700/3.

3.1.2 Arm support post and anchor block. The anchor block and arm support post shall be aluminum conforming to alloy 2024-T4 of QQ-A-225/6. As an alternate, the support post and anchor block shall be 6061 aluminum rod conforming to QQ-A-225/8, Temper T6.

3.1.3 Arm support. The arm support shall be corrosion resisting stainless steel conforming to class 301 of QQ-S-766. As an alternate, the arm support shall be aluminum conforming to alloy 5052-H32 of QQ-A-250/8.

3.1.4 Handle casting. If cast, the hand grip handle shall be either aluminum alloy sand casting NO. 295.0, No. 356.0, or 713 conforming to QQ-A-601. Temper shall be T-6 unless otherwise indicated. As an alternate, aluminum die casting alloy No. 380 conforming to requirements of QQ-A-591 (including soundness and workmanship) shall be used.

3.1.5 Handle forging. If forged, the hand grip handle shall be aluminum alloy conforming to QQ-A-367, composition 2014-T6, or equal.

3.1.6 Plastic and rubber. The crutch tip shall be made of a compound of natural rubber, synthetic rubber or a mixture of the two. Color of the tips shall be maroon, natural, or other light color. The hand grip cover shall be plastic polyvinyl chloride and shall conform to the properties listed in table I. Color shall match the tip.

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TABLE I. Properties of vinyl hand grip.

Characteristic	Requirement
Specific gravity	1.30 to 1.40
Water absorption (24 hr), percent, maximum	0.5
Flammability	self-extinguishing
Tensile strength, psi, minimum	1,850
Elongation, percent (in 2 in.)	300 to 400
Hardness (Shore A)	70 to 80

3.2 Construction. The crutch shall be of single, telescoping aluminum tube construction with projecting plastic grip covered handle, and a C-shaped corrosion-resisting steel or aluminum arm support. Bottom of crutch shall be fitted with a heavy rubber crutch tip. Construction details shall conform to the drawing. Starting above hand grip, tubing shall bend slightly to provide comfortable positioning of arm support, which shall be attached by means of an anchor block to an adjusting post telescoping into the upper end of tubing. Adjusting post shall be adjustable to four different heights. The lower portion of the crutch shall also be adjustable in length, to eight positions. A 300 pound weight shall be borne by the crutch without damage. Shall conform to Defense Personnel Support Center Drawing 1367.

3.2.1 Upper tube section of crutch. Upper tube section of crutch shall be formed cold, without wrinkling interior of bend. Holes for tapered head screws must be jig drilled and reamed to match taper of mating parts to not less than a class 2 fit.

3.2.2 Lower tube section. Lower tube section shall have a smooth telescopic fit. Tube shall be jig drilled for interchangeability from side to side and between holes, and shall be jig reamed to not less than a class 2 fit with matching tapered holes in upper tube.

3.2.3 Locking screws and tapered sleeves. Locking screws and tapered sleeves shall be machined to a class 2 or better fit with tapered holes, plus tolerance only on tapers to compensate for wear of the aluminum tubing.

3.2.4 Elastic stop nuts. Corrosion-resisting steel stop nuts for locking screws shall be the size shown on Drawing 1367. Countersunk flathead bolt for hand grip and stop nuts shall be machined to countersink taper as shown on the drawing.

3.2.5 Arm support. Arm support shall be cold formed with all sharp edges broken as shown on the drawing. Dimples for countersinks shall be of sufficient depth to allow heads of mating screws to set nearly flush with interior curvature of arm support when screwed into anchor block.

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3.2.6 Anchor block. Countersinks in block shall be machined to fit dimples in arm support band as closely as possible without breaking through side wall of block.

3.2.7 Arm support adjusting post. Arm support adjusting post shall be drilled and reamed to match upper tube section mating holes, side for side, to at least a class 2 fit. Post shall be machined for a sliding fit in upper tube section. Very slight countersink is indicated for this reason, and heading allowance shall be kept at a minimum to prevent cracking of the aluminum alloy rivets. Riveting shall be accomplished so that little or no resistance is offered by shank of rivet to the rotation of the arm support block.

3.2.8 Hand grip handle. The hand grip metal shall be free of pits, blowholes and other casting or forging defects. The metal hand grip shall be machined so that all hand grips will be interchangeable with all crutches and shall have a sliding fit on upper tube section for replacement purposes. Grip handle shall present a smooth surface before sliding on the plastic hand grip cover. The hand grip cover shall cover handle as shown on the drawing and shall fit sufficiently tight to prevent slippage in use.

3.2.8.1 Hand grip cover. The design and dimensions of the hand grip cover shall be as shown on the drawing. All flash shall be removed.

3.2.8.2 Bolt. The bolt used to secure the hand grip handle to the crutch tubing shall be a flat-head bolt. The bolt shall be machined as shown on the drawing to a class 2 fit for this thread type. Shank shall be capable of bridging upper tube section and hand grip without contact with bolt threads. The bolt shall be secured with the elastic stop nut.

3.2.9 Crutch tip. The crutch tip shall be standard 7/8 inch with hole reamed to fit tubing, and shall conform to details of the drawing. A metal bearing plate shall be molded into the tip. Hardness of the rubber tip shall be 40 to 55, Shore Durometer "A". As an alternate to the rubber, the crutch tip shall be fabricated of a plastic material, suitable for the purpose intended, firmly bonded to the metal. The crutch tip shall be abrasion resistant, and non marking when tested as described in 4.5.2 and 4.5.3.

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3.3 Finish.

3.3.1 Aluminum. Parts shall be anodized in accordance with type II of MIL-A-8625, to a uniformly dense oxide coating. As an alternate, hand grip casting or forging shall be machined smooth and given a chemical coating conforming to MIL-C-5541, class 1A.

3.3.2 Corrosion-resisting steel. Parts shall receive no additional finishing beyond machine finish acceptable in screw machine production.

3.4 Identification marking. Each crutch shall be permanently and legibly marked with the manufacturer's name or registered trademark, placed in an inconspicuous location.

3.5 Workmanship. Workmanship shall be first class throughout. Crutches shall be well made, finished in appearance, and free from any defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

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4.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.2.1 Certificates of quality. Certificates of quality, supplied by the manufacturer of the material and components may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the date of testing, the test method, and individual test results.

4.3 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered for acceptance are classified as specified in tables II and III (see 4.3.1 and 4.4.1).

4.3.1 Sampling for tests and examination. Sampling for tests and examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels shall be as indicated in table II. Unit of product for examination and test purposes shall be one crutch.

TABLE II. Sampling plans.

For tests	Inspection level	AQL
End item and component	S-1 1/	2/
For examination		
For defects	I	Major 1.0 Minor 2.5

1/ A minimum of 3 test samples shall be used.
2/ Acceptance number shall be zero.

4.3.2 For chemical analysis of the metals. Four ounces of each lot of each metal used in the construction of the items shall be selected for chemical analysis as specified in 4.5.2.

4.4 Examination. The crutch shall be examined to determine compliance with all requirements contained in this specification.

4.4.1 Classification of defects. Examination shall be conducted in accordance with, but not restricted to, the classification of defects listed in table III. Additional defects shall be classified in accordance with MIL-STD-105.

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TABLE III. Categories and defects

Major:	
101	Crutch not complete with all component parts.
102	Components not properly assembled or aligned.
103	Exposed and accessible edges and surfaces not free from sharp edges and burrs.
104	Dimensions not within tolerances specified.
Minor:	
201	Design and construction not as shown on the drawing.
202	Aluminum parts not treated as specified.
203	Manufacturer's name or trademark not located as specified, illegible, or omitted.

4.4.2 Packaging inspection. The inspection of the unit package, packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.5 Tests. Tests shall be conducted to determine compliance with specification requirements. Where possible, the same sample unit shall be used for the determination of two or more test characteristics.

4.5.1 Test methods. Test methods shall be as specified herein and in the applicable subsidiary specifications and in table IV for the hand grip plastic.

TABLE IV. Test methods for vinyl plastic

Test	ASTM Method
Specific gravity	D 792
Water absorption	D 570
Flammability	D 635
Tensile strength	D 412
Elongation	D 638
Hardness	D 2240

4.5.2 Chemical analysis of the metals. The samples selected in accordance with 4.3.2 shall be analyzed in conformance with methods 111 or 112 of Federal Test Method Standard No. 151. In case of dispute, method 111 shall be used for referee analysis.

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4.5.3 Crutch tip resistance to abrasion. The resistance to abrasion of the tips shall be determined in method 14111 of Fed. Test Method Std. No. 601.

4.5.4 Crutch tip marking. One tip from each test unit shall be exposed to air at elevated temperature as described in method 7221 or 7231 of Fed. Test Method. Std. No. 601. At the end of the exposure period, the exposed tip shall be examined for tackiness and brittleness. The edge of the wearing surface of the exposed tip shall be drawn across a sheet of white paper. If only a slight mark which is easily rubbed off with the fingers is made on the paper, the tip shall be considered as nonmarking.

5. PACKAGING

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit package. Each crutch shall be packaged in a box of appropriate size constructed in accordance with PPP-B-636, type CF, class domestic. Box shall be closed as specified in the appendix of PPP-B-636.

5.1.2 Level C. Crutches shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be Level A, B or C as specified (see 6.2).

5.2.1 Level A. Sixteen (16) crutches shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed as specified in the appendix of PPP-B-636.

5.2.2 Level B. Sixteen (16) crutches shall be packaged in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Closure of fiberboard boxes shall conform to method II as specified in PPP-B-636.

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5.2.3 Level C. Crutches shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking. Each unit package and exterior container shall be marked as specified in MIL-STD-129.

6. NOTES

6.1 Intended use. The crutch covered by this specification is intended for use as an aid in ambulation.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable level of packaging and packing (see 5.1 and 5.2).
(c) National stock number.

6.3 This specification is intended to cover the following item appearing in the Federal Supply Catalog:

National Stock Number	Item Identification
6515-00-777-7315	CRUTCH CANE TYPE, Adjustable

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6515-2013

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-C-16671B - CRUTCH, ALUMINUM, CANE TYPE, ADJUSTABLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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